Transport model

Technical derivation

⚠ Boring Math Stuff

In a co-moving frame, the classical Hamiltonian describing transport of a mixed-species N-ion chain is

$$H = \frac{1}{2} p^T M^{-1} p + V(t) - p^T \frac{\partial \bar{q}(t)}{\partial t}$$

Ansatz: the total force on the ions is approximately

$$-\nabla_q V(t) = -K(t)q + \eta(t)$$

where $K(t)$ is the potential Hessian and $\eta(t)$ is a random force representing an error in the applied potential.

Symbol	Definition
$\bar{q}(t) \in \mathbb{R}^{3N}$	Equilibrium positions
$q \in \mathbb{R}^{3N}$	Ion displacements from equilibrium
$p \in \mathbb{R}^{3N}$	Ion momenta
$M \in \mathbb{R}^{3N \times 3N}$	Ion masses (diagonal real-valued matrix)
$V(t)$	Potential energy
$K(t) \in \mathbb{H}^{3N}$	Hessian of the potential @ $\bar{q}(t)$
$\eta(t) \in \mathbb{R}^{3N}$	Random force (error) acting on ions

Error sources:

- DC stray fields,
- Control errors in applied voltages,
- Distortions from DC filters,
- Small changes in impedance on interior traces in the chip, vacuum interconnects, or electrical components.

Key research question:

Given an error model and bound for $\eta(t)$, what do they imply for the architecture's allowable heating budget, achievable transport speed, and required ion cooling?

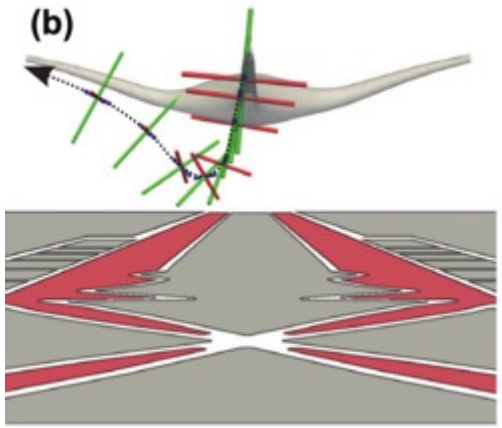
Transport model

Technical derivation

⚠ Boring Math Stuff

Normal modes:

- Normal modes are local to each point along the transport waveform.
- A global normal-mode basis offers little computational benefit, since the modes change along the trajectory.
- We only need the normal-mode basis at the end of transport, when computing phonon occupations.



Basic idea:

- Time-independent transformation to mass-weighted coords.
- Chosen to correspond to normal mode basis at the end of the end of the transport interval $t \in [0, \tau]$.

$$\tilde{K}(t) = M^{-1/2}\, K(t)\, M^{-1/2}$$

$$\Omega^2(t) = S^T\, \tilde{K}(t)\, S$$

S is the orthogonal matrix such that $\Omega^2(\tau)$ is diagonal

Define:

$$\Omega_\tau = \Omega(\tau) \qquad Q = \Omega_\tau^{1/2} S^T M^{1/2} q \qquad P = \Omega_\tau^{-1/2} S^T M^{-1/2}\, p$$

Symbol	Definition
$\tilde{K}(t) \in \mathbb{H}^{3N}$	Mass weighted Hessian
$S \in O(3N)$	Orthogonal matrix that diagonalizes $\tilde{K}(t)$ @ end of transport τ
$p \in \mathbb{R}^{3N}$	Ion momenta
$\Omega^2(t) \in \mathbb{H}^{3N}$	Image of $\tilde{K}(t)$ under transf. by S
Q	Generalized position
P	Generalized momenta

[1] Wright et al., NJP (15) 033004 2013

Transport model

Technical derivation

⚠ Boring Math Stuff

In terms of the quadrature coordinates, the Hamiltonian is

$$H = \frac{1}{2}\left(P^T \Omega_\tau P + Q^T\, \widetilde{\Omega}(t)\, Q\right) - Q^T \zeta(t) - P^T \gamma(t)$$

Displacements due to errors in $V(t)$

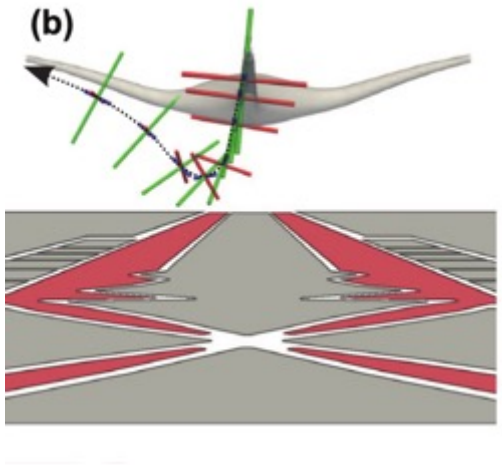
Displacements induced by co-moving frame

where

$$\widetilde{\Omega}(t) = \Omega_\tau^{-1/2} \Omega^2(t)\, \Omega_\tau^{-1/2}$$

$$\gamma(t) = \Omega_\tau^{1/2}\, S^T M^{1/2}\, \left(\partial \overline{q}/\partial t\right)$$

$$\zeta(t) = \Omega_\tau^{-1/2} S^T M^{-1/2}\, \eta(t)$$



Hamilton's equations can be rewritten as an affine stochastic system

$$\dot{x} = A(t)x + B\zeta(t) + C(t)$$

$$x = \begin{pmatrix} Q \\ P \end{pmatrix} \qquad A(t) = \begin{pmatrix} 0 & \Omega_\tau \\ -\widetilde{\Omega}(t) & 0 \end{pmatrix} \qquad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes I_{3N} \qquad C(t) = -\begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \gamma(t).$$

Transport model

Technical derivation

⚠ Boring Math Stuff

In the absence of errors, the E.O.M. are deterministic and the **ideal** trajectory is

$$x_0(t) = \Phi(t,0)x(0) + \int_0^t \Phi(t,s)C(s)\,ds$$

where

$$\frac{\partial}{\partial t}\Phi(t,s) = A(t)\Phi(t, s), \qquad \Phi(s,s) = I_{6N}\,.$$

Symbol	Definition
$x_d(t) \in \mathbb{R}^{6N}$	Ideal (error free) phase-space vector
$x(t) \in \mathbb{R}^{6N}$	Actual phase-space vector
$x_e(t) \in \mathbb{R}^{6N}$	Deviation from the ideal phase-space vector caused by random forces
$\Phi(t,s) \in \mathbb{R}^{6N\times 6N}$	Fundamental matrix solution
$\Sigma_x(t) \in \mathbb{R}^{6N\times 6N}$	Phase space covariance matrix

Random forces displace the ions from the ideal trajectory. Including these forces the actual trajectory is

$$x(t) = x_0(t) + x_e(t)$$

Usually made 0 as the objective of waveform design

$$x_e(t) = \int_0^t \Phi(t,s)\,B\,\zeta(s)\,ds$$

Nonzero due to errors.

$$\Sigma_x(t) = \mathbb{E}[x_e(t)x_e^T(t)]$$

Covariance matrix characterizes distribution in phase space.

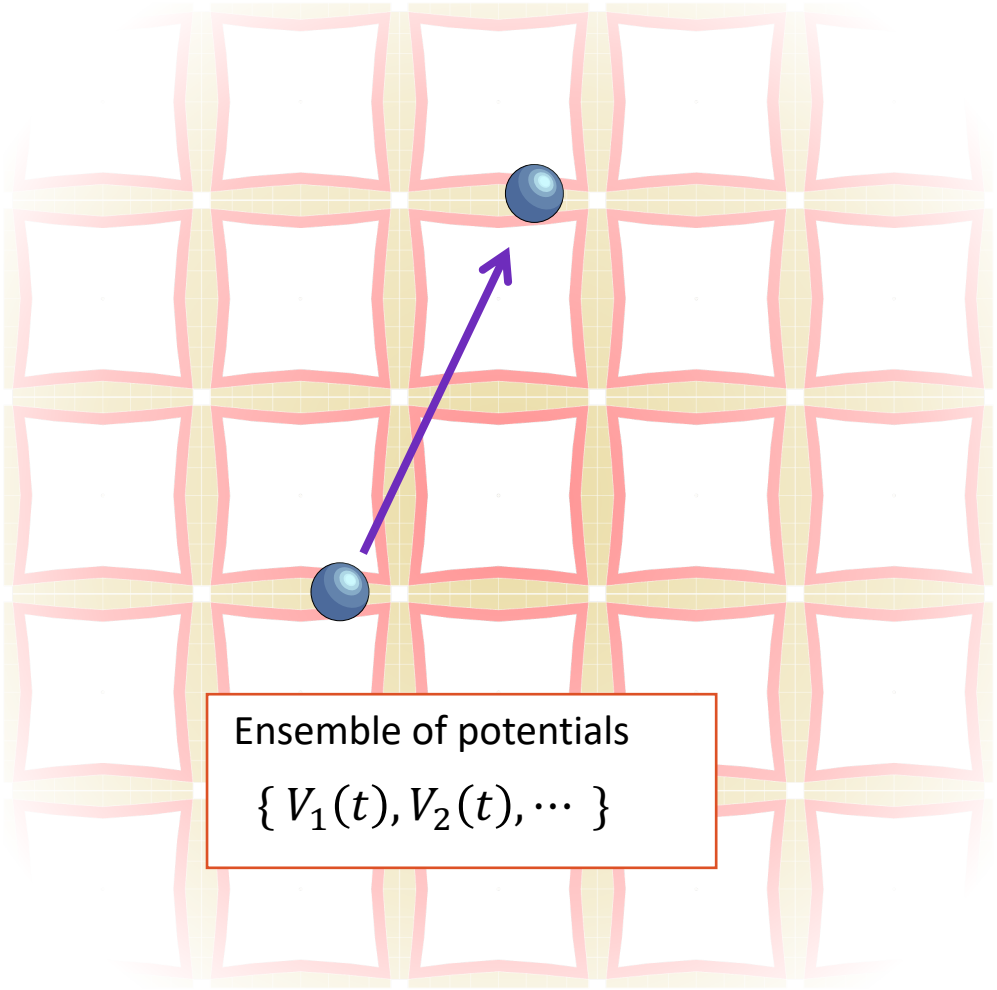
Broadcasting ensemble

- The use of broadcasting implies a translational symmetry.
- This symmetry produces a lattice of of moving wells with similar, though not identical, potentials.
- The broadcasting ensemble is the statistical ensemble of these independently evolving wells.

<u>Potential noise leads to spread in phase space</u>

$$G(s,s') = \mathbb{E}[\zeta(s)\zeta^T(s')]$$

Average over the broadcasting ensemble

The system covariance $\Sigma_x(t)$ is the accumulated effect of time-correlated noise, propagated into phase space by $\Phi(t,s)$

$$\Sigma_x(t) = \int_0^t \int_0^t \Phi(t,s)\, B\, G(s,s')\, B^T \Phi^{\mathrm{T}}(t,s')\, ds\, ds' \qquad (1)$$